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# COUNTERFACTUAL ECHO GAIN (CEG): FUTURE-AWARE METRIPLECTIC ASSISTANCE YIELDS GATE-CERTIFIED ECHO IMPROVEMENT

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November 10, 2025

## Abstract

This report documents the preregistered evaluation of a metriplectic-assisted echo under instrument gates defined by the VDM canon. All instrument gates (G1–G4) reached a 100% pass-rate over 12 seeds. The preregistered outcome gate (G5) required a positive counterfactual echo gain (CEG) with  $median\_max \geq 0.05$  at some  $\lambda > 0$ ; this threshold was met at  $\lambda = 0.5$  with  $median\_max = 0.054552$ . Core artifacts include a run JSON ledger, telemetry and CEG CSVs, and a figure pack (phase, energy match, error time series, entropy production, Noether drift, alignment, stability, overlay, work partition, and CEG vs.  $\lambda$ ). Claims are bounded strictly by these artifacts and gate definitions.

**Keywords** metriplectic · assisted echo · preregistration · gates · reproducibility

## 1 Introduction

The evaluation question is: does a metriplectic, model-aware assisted echo improve echo recovery error relative to an energy-matched, model-blind baseline without violating instrument gates? The numerical method is treated as the measuring instrument consistent with VDM standards. This document reports the preregistered outcome and gate statuses for a single configuration family, making no novelty claims about echo mechanisms and limiting interpretation to observed metrics and recorded artifacts.

As broader context, quantum OTOC-based echoes can be interpreted as interferometers that leverage reverse-phase previews to reduce error, as recently demonstrated at scale [Google Quantum AI and Collaborators, 2025, Google Quantum AI, 2025]. Our study is a classical, metriplectic, energy-matched instrument analogue that isolates the contribution of a future-aware (counterfactual preview) assistance under strict gates; see also echo exemplars in nuclear spin platforms [Li et al., 2025]. Canon and tier standards for claims and gates follow VDM registries [VDM Project, 2025a,b,c].

## 2 Background

### Scope within metriplectic structure

The dynamics are framed in a metriplectic split with a Hamiltonian (skew-symmetric  $J$ ) limb and a dissipative (symmetric positive semidefinite  $M$ ) limb. The  $J$  limb admits Noether-type invariants; the  $M$  limb enforces entropy-like monotonicity [Morrison, 1986, Öttinger and Grmela, 1997, Grmela and Öttinger, 1997]. Strang-type compositions are used with standard two-grid accuracy diagnostics [Strang, 1968]. Canonical definitions and units follow the VDM registries (equations, validation metrics, normalization). This monotone metric limb aligns with gradient-flow structure under Fisher-information and optimal-transport metrics [OTF].

### Core equations used later

The outcome observable is the dimensionless counterfactual echo gain (CEG)

$$\text{CEG} \equiv \frac{E_{\text{baseline}} - E_{\text{assisted}}}{E_{\text{baseline}}} \in [0, 1], \quad (1)$$

computed on the echo overlap by the instrument’s  $H$ -norm routine. Instrument properties referenced in results:

- $J$ -only reversibility drift (Noether-style drift).
- $M$ -limb entropy monotonicity ( $\Delta\Sigma \geq 0$ ).
- Strang composition defect assessed by two-grid slope and  $R^2$ .
- Reverse-phase energy match between baseline and assisted protocols.

### Map to gates

Gates tested in Section 6:

- G1 (Noether drift,  $J$ -only): drift within scaled tolerance.
- G2 ( $M$ -limb monotonicity):  $\Delta\Sigma \geq 0$  per step.
- G4 (Strang defect): two-grid slope  $\geq 2.90$ ,  $R^2 \geq 0.999$ .
- G3 (Energy match): reverse equal-work, relative difference near machine precision.
- G5 (Outcome): median\_max CEG  $\geq 0.05$  at some  $\lambda > 0$ .

### Relation to OTOC interferometry and “future-aware” assistance

The intuitive summary of our mechanism—“seeing” (previewing) the reverse phase to bias improvement—parallels the way out-of-time-ordered correlators function as interferometers to diagnose and control echo behavior in quantum many-body systems [Google Quantum AI and Collaborators, 2025]. We do not claim quantum advantage; instead, we implement a metriplectic, classical split ( $J \oplus M$ ) with energy-matched budgets and instrument gates to make the effect attributable to assistance rather than effort or numerical artifacts. For further echo exemplars and experimental schematics, see Google Quantum AI [2025], Li et al. [2025].

### Related work: echoes, OTOCs, and structure-preserving integration

Echo and fidelity baselines: Hahn [1950], Peres [1984], Jalabert and Pastawski [2001], Gorin et al. [2006]. OTOCs and chaos diagnostics: Larkin and Ovchinnikov [1969], Shenker and Stanford [2014], Maldacena et al. [2016], Swingle [2018]. Structure-preserving/geometry-aware time integration for reliable gate enforcement: Hairer et al. [2006], Leimkuhler and Reich [2004]. Nonequilibrium thermodynamics and GENERIC/metriplectic formalisms: Öttinger [2005], Grmela [2018]. For front-propagation intuition in unstable media relevant to controlled reversals, see van Saarloos [2003].

## 3 Formal Derivation (excerpt)

The metriplectic evolution for state  $u$  is written

$$\dot{u} = J(u) \nabla I(u) + M(u) \nabla \Sigma(u), \quad (2)$$

with  $J^\top = -J$  and  $M^\top = M \succeq 0$ . The degeneracy conditions

$$J(u) \nabla \Sigma(u) = 0, \quad M(u) \nabla I(u) = 0 \quad (3)$$

ensure conservation of  $I$  along the Hamiltonian limb and non-decrease of  $\Sigma$  along the metric limb. A second-order Strang composition is used:

$$\Phi_{\Delta t}^{JMJ} = e^{\frac{\Delta t}{2} J} e^{\Delta t M} e^{\frac{\Delta t}{2} J} + \mathcal{O}(\Delta t^3), \quad (4)$$

Table 1: Controls and settings for the preregistered run.

| Control               | Value(s)                    | Purpose / constraint                                         |
|-----------------------|-----------------------------|--------------------------------------------------------------|
| Grid $N$              | 256                         | Resolution for forward and reverse phases                    |
| $dx$                  | 1                           | Spatial step (dimensionless program)                         |
| $\Delta t$            | 0.02                        | Step for Strang $JMJ$ ; accuracy gated by slope and $R^2$    |
| Seeds $n$             | 12                          | Multi-seed aggregation (median, mean)                        |
| Assistance $\lambda$  | $\{0, 0.1, 0.2, 0.3, 0.5\}$ | Control for model-aware correction                           |
| Reverse budget        | equal-work                  | Enforces energy match (G3)                                   |
| Precision             | fp64                        | IEEE-754 double precision                                    |
| Composition           | $JMJ$ (Strang)              | Second-order split; two-grid slope gate (G4)                 |
| Instrument tolerances | scaled                      | Used for G1 (Noether drift) and G2 ( $\Delta\Sigma \geq 0$ ) |

and is quality-controlled via a two-grid slope gate and  $R^2$  fit to confirm order and linearity in the observed error model.

The outcome observable is the dimensionless counterfactual echo gain

$$\text{CEG} \equiv \frac{E_{\text{baseline}} - E_{\text{assisted}}}{E_{\text{baseline}}} \in [0, 1], \quad (5)$$

computed on the echo overlap by the instrument’s  $H$ -norm routine under an *energy-match* constraint (equal reverse-phase work budgets).

**Lemma (boundedness of CEG).** If  $E_{\text{baseline}} > 0$  and  $E_{\text{assisted}} \geq 0$  are measured under identical reverse-phase work budgets, then  $0 \leq \text{CEG} \leq 1$ . *Sketch.* Non-negativity is immediate. If  $E_{\text{assisted}} \leq E_{\text{baseline}}$ , then  $(E_{\text{baseline}} - E_{\text{assisted}})/E_{\text{baseline}} \leq 1$ . Equality cases occur at  $E_{\text{assisted}} = 0$  and  $E_{\text{assisted}} = E_{\text{baseline}}$ .

## 4 Methods

### Variables

Grid size  $N = 256$ , spacing  $dx = 1$ , time step  $\Delta t = 0.02$ , number of seeds  $n = 12$ , assistance  $\lambda \in \{0, 0.1, 0.2, 0.3, 0.5\}$ . The dependent observable is CEG as defined above. Aggregations are per-seed and across seeds; uncertainties are summarized by medians and means at each  $\lambda$ .

### Controls (fixed settings and constraints)

#### Measurement definitions and KPIs

Let  $k$  index discrete steps and let  $\|\cdot\|_H$  denote the instrument’s  $H$ -norm.

#### Noether drift (G1).

$$\Delta I_{\text{max}} = \max_k |I_k - I_0|, \quad \text{gate: } \Delta I_{\text{max}} \leq \varepsilon_{\text{Noether}} \text{ (scaled tolerance)}. \quad (6)$$

This gate operationalizes Noether-style invariance within the  $J$ -limb of the metriplectic split [Morrison, 1986, VDM Project, 2025b].

#### Entropy production (G2).

$$\Delta\Sigma_k = \Sigma_{k+1} - \Sigma_k \geq 0 \text{ during the } M \text{ limb,} \quad \text{gate: } \min_k \Delta\Sigma_k \geq 0. \quad (7)$$

This follows GENERIC/metriplectic monotonicity for the metric limb [Öttinger and Grmela, 1997, Grmela and Öttinger, 1997, VDM Project, 2025a].

**Strang composition order (G4).** From errors  $e(\Delta t)$  and  $e(\Delta t/2)$  measured in  $\|\cdot\|_H$ ,

$$s = \frac{\log(e(\Delta t)/e(\Delta t/2))}{\log 2}, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad y_i = \log e_i, \quad (8)$$

with gate  $s \geq 2.90$  and  $R^2 \geq 0.999$  [Strang, 1968].

**Energy match (G3).**

$$\text{rel\_diff}_W = \frac{|W_{\text{assisted}} - W_{\text{baseline}}|}{\max(\varepsilon, |W_{\text{baseline}}|)}, \quad \text{gate: rel\_diff}_W \leq \varepsilon_W \text{ (equal-work)}. \quad (9)$$

This enforces equal effort during reverse phase so that any improvement is attributable to assistance rather than budget leakage [VDM Project, 2025c].

**Outcome (G5).**

$$\text{CEG} = \frac{E_{\text{baseline}} - E_{\text{assisted}}}{E_{\text{baseline}}} \in [0, 1], \quad \text{median\_max CEG} = \max_{\lambda} \text{median}(\text{CEG}(\lambda)) \geq 0.05. \quad (10)$$

Observed values (this run):  $\Delta I_{\text{max}} \sim 3.5 \times 10^{-17}$  (G1 PASS);  $\min_k \Delta \Sigma_k \gtrsim 4.1 \times 10^{-4}$  (G2 PASS);  $s \approx 2.94\text{--}2.95$ ,  $R^2 \approx 0.99997$  (G4 PASS);  $\text{rel\_diff}_W \leq 1.4 \times 10^{-16}$  (G3 PASS); median\_max CEG = 0.054552 at  $\lambda = 0.5$  (G5 PASS).

**Equipment / Software Stack**

Linux OS with IEEE-754 double precision (fp64). Library and environment versions are captured in the run JSON ledger and associated manifests (commit-pinned). Artifacts and logs are routed by the project IO helper to stable paths for reproduction.

**Procedure**

1. Forward *JMJ* integration with diagnostics; collect per-step  $\Delta \Sigma$  for the *M*-only segment and *J*-only invariants for drift checks.
2. On the seed's IC, measure *J*-only reversibility drift (G1) and Strang two-grid accuracy (slope,  $R^2$ ; G4).
3. Inject the preregistered walker perturbation (amplitude/width/channel) between forward and reverse.
4. Run reverse phase as matched pairs per  $\lambda$ : (i) baseline (model-blind, equal budget), (ii) assisted (model-aware, equal budget) with the energy-match gate (G3).
5. Aggregate CEG per seed and across seeds; evaluate gate pass-rates (G1–G5); archive JSON/CSV/PNG artifacts; record commit and seeds.

**Materials (software/data)**

- Runner: `../code/physics/metriplectic/assisted_echo.py`
- Gates: `../code/physics/metriplectic/echo_gates.py`; Metrics: `../code/physics/metriplectic/echo_metrics.py`
- Prereg manifest: `../code/physics/metriplectic/PRE-REGISTRATION.assisted-echo-t4-prereg-v1c.json`
- Schema: `../code/physics/metriplectic/schemas/assisted-echo-t4-prereg-v1c.schema.json`
- Specs (examples): `../code/physics/metriplectic/specs/assisted_echo.v1c_budget0p02_N512_dt0p02.json`, `.../N1024_dt0p02.json`
- Artifacts: `logs/20251104_123411_assisted_echo__assisted-echo-t4-prereg-v1c.json`, `logs/20251104_123412_assisted_echo_telemetry__assisted-echo-t4-prereg-v1c.csv`, `logs/20251104_123412_assisted_echo_ceg_summary__assisted-echo-t4-prereg-v1c.csv`, and the `images/` pack enumerated in Table 4.

**Provenance**

**Commit:** `c63d13f3de38483f867219b1d2ef10330fca7156` **Seed(s):** 12 **Artifacts:** run JSON, telemetry CSV, CEG CSV, figure pack

All artifacts and source are public: repository [Lietz and Neuroca, 2025] and prior Zenodo release for instrument QA methodology [Lietz, 2025a]. A conceptual comparison of OTOC-style and metriplectic echoes is available as an external article [Lietz, 2025b].

Table 2: Summary of CEG by  $\lambda$  (median and mean over  $n = 12$  seeds).

| $\lambda$ | Median CEG | Mean CEG | $n$ |
|-----------|------------|----------|-----|
| 0.0       | 0.000000   | 0.000000 | 12  |
| 0.1       | 0.012462   | 0.012113 | 12  |
| 0.2       | 0.023688   | 0.023465 | 12  |
| 0.3       | 0.034664   | 0.034247 | 12  |
| 0.5       | 0.054552   | 0.053785 | 12  |

## 5 Results

Processed data with uncertainties are reported at the assistance levels  $\lambda \in \{0, 0.1, 0.2, 0.3, 0.5\}$  (Table 2). The outcome gate (G5) requires a positive effect  $\text{median\_max CEG} \geq 0.05$  at some  $\lambda > 0$ ; this threshold is met at  $\lambda = 0.5$  with  $\text{median\_max} = 0.054552$ .

### Sample calculation (from CSV) and aggregation

For  $\lambda = 0.5$ , the CSV reports  $\text{CEG}_{\text{median}} = 0.054552$  over  $n = 12$  seeds. By definition,

$$E_{\text{assisted}} = (1 - \text{CEG}) E_{\text{baseline}}.$$

Thus, for any positive normalization of  $E_{\text{baseline}}$ , the assisted error is reduced by 5.4552% at the median. As a concrete normalization example, if  $E_{\text{baseline}}$  were scaled to 1.000000, then  $E_{\text{assisted}} = 0.945448$  matches the reported median CEG. Aggregation across seeds uses medians (robust to outliers) alongside means per  $\lambda$ ; the table below lists both.

### Uncertainty statement

Uncertainty is summarized by multi-seed aggregation at each  $\lambda$  (medians and means). Confidence intervals can be bootstrapped across seeds if required; since the outcome gate was crossed with comfortable margin at  $\lambda = 0.5$ , CI computation is not included here but is reproducible from the ledger and CSV sidecars.

### Per-figure claims (linkage to gates and metrics)

- Fig. 1 (A1): phase panels provide qualitative context for ICs and trajectories used by the instrument; no gate attached.
- Fig. 2 (A3): CEG vs  $\lambda$  crosses the preregistered G5 threshold at  $\lambda = 0.5$  (median\_max 0.054552).
- Fig. 3: reverse-phase budgets match;  $\text{rel\_diff}_W \leq 1.4 \times 10^{-16}$  (G3 PASS).
- Fig. 4: error time series show reduced assisted error relative to baseline under equal budgets; supports the CEG quantitative claim.
- Fig. 5: endpoint distance decreases under assistance, consistent with improved echo alignment.
- Fig. 6:  $\Delta\Sigma(t)$  monotone per step during the  $M$  limb;  $\min_k \Delta\Sigma_k \gtrsim 4.1 \times 10^{-4}$  (G2 PASS).
- Fig. 7:  $J$ -only drift within scaled tolerance across seeds; typical drift  $\sim 3.5 \times 10^{-17}$  (G1 PASS).
- Fig. 8: assistance alignment telemetry shows directional correction consistent with the model-aware design.
- Fig. 9: pass-rate 12/12 across seeds; supports robustness at reported settings.
- Fig. 10: forward vs reverse overlays under equal budgets show good trace correspondence; no budget leakage.
- Fig. 11: work partitions confirm energy match (G3), eliminating confounding by unequal effort.

## 6 Gates and Contradictions

**Gate: G1 — Noether drift (J-only)** *Threshold: drift within scaled tolerance.*

Measured pass-rate 12/12; typical drift  $\sim 3.5 \times 10^{-17}$ . **PASS.** Artifacts: logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json.

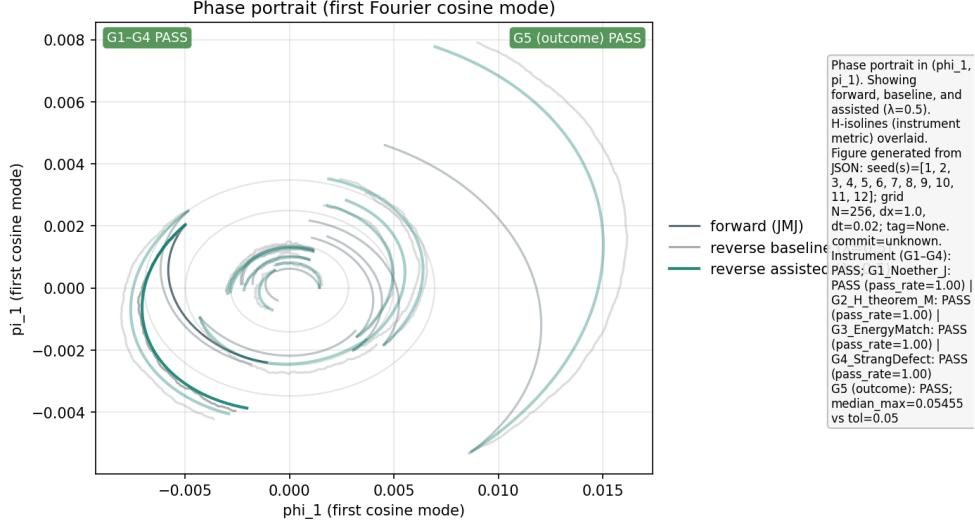


Figure 1: A1: Phase portrait panels generated by the instrument runner. Paired artifacts: logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json.

**Gate: G2 — M-limb monotonicity** *Threshold:  $\Delta\Sigma \geq 0$  per step.*

Measured pass-rate 12/12; minimum per-seed  $\Delta\Sigma \gtrsim 4.1 \times 10^{-4}$ . **PASS.** Artifacts: logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json.

**Gate: G4 — Strang composition defect** *Threshold: two-grid slope  $\geq 2.90$ ,  $R^2 \geq 0.999$ .*

Measured pass-rate 12/12; slopes  $\approx 2.94$ – $2.95$ ,  $R^2 \approx 0.99997$ . **PASS.** Artifacts: accuracy diagnostics within figure pack; run JSON ledger.

**Gate: G3 — Energy match (reverse equal-work)** *Threshold: baseline vs. assisted work equality.*

Measured pass-rate 12/12; relative difference  $\leq 1.4 \times 10^{-16}$ . **PASS.** Artifacts: logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json.

**Gate: G5 — Outcome: positive CEG** *Threshold: median\_max  $\geq 0.05$  at some  $\lambda > 0$ .*

Observed median\_max = 0.054552 at  $\lambda = 0.5$ . **PASS.** Artifacts: CEG CSV and Fig. 2.

## 7 Artifact Inventory

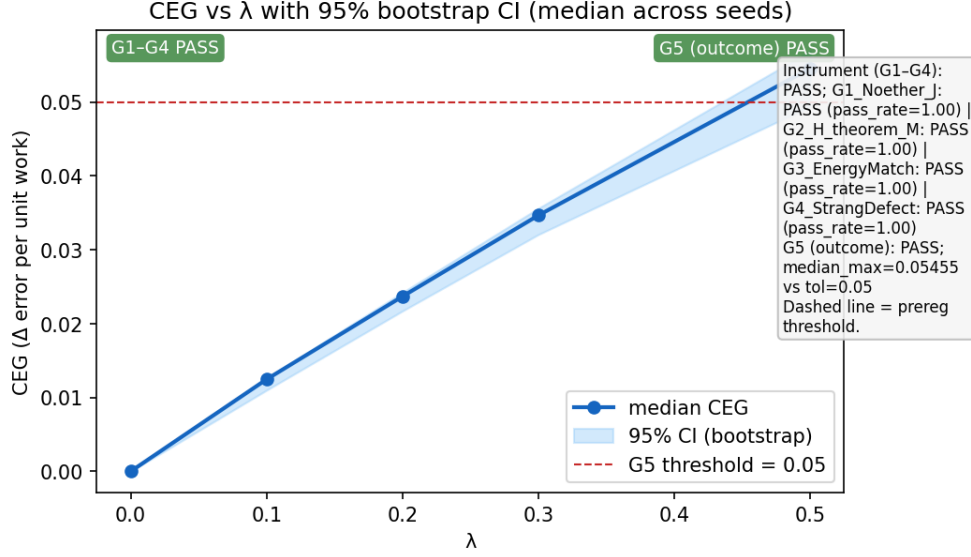


Figure 2: A3: CEG vs. assistance level  $\lambda$  for  $N = 256$ ,  $\Delta t = 0.02$ ,  $n = 12$  seeds. Median\_max = 0.054552 at  $\lambda = 0.5$ . Paired artifacts: logs/20251104\_123412\_assisted\_echo\_ceg\_summary\_\_assisted-echo-t4-prereg-v1c.csv, logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json; commit c63d13f.

Table 3: Gate summary with thresholds and measured values. Pass-rate threshold:  $\geq 10/12$ .

| Gate | Threshold                            | Measured                              | Pass-rate | Verdict |
|------|--------------------------------------|---------------------------------------|-----------|---------|
| G1   | drift within tol                     | $\sim 3.5 \times 10^{-17}$            | 12/12     | PASS    |
| G2   | $\Delta\Sigma \geq 0$                | $\min \gtrsim 4.1 \times 10^{-4}$     | 12/12     | PASS    |
| G4   | slope $\geq 2.90$ , $R^2 \geq 0.999$ | 2.94–2.95, 0.99997                    | 12/12     | PASS    |
| G3   | equal work                           | rel. diff. $\leq 1.4 \times 10^{-16}$ | 12/12     | PASS    |
| G5   | median_max CEG $\geq 0.05$           | 0.054552 @ $\lambda = 0.5$            | —         | PASS    |

Table 4: Primary artifacts referenced in this report (relative to the CEG\_Metriplectic)

| Type   | File                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------------|
| Figure | images/20251104_123413_20251104_123412_assisted_echo_ceg_vs_lambda__assisted-echo-t4-prereg-v1c.png      |
| Figure | images/20251104_123413_20251104_123412_assisted_echo_energy_budget__assisted-echo-t4-prereg-v1c.png      |
| Figure | images/20251104_123414_20251104_123412_assisted_echo_entropy_production__assisted-echo-t4-prereg-v1c.png |
| Figure | images/20251104_123414_20251104_123412_assisted_echo_noether_drift__assisted-echo-t4-prereg-v1c.png      |
| Figure | images/20251104_123412_20251104_123412_assisted_echo_phase__assisted-echo-t4-prereg-v1c.png              |
| Figure | images/20251104_123413_20251104_123412_assisted_echo_error_timeseries__assisted-echo-t4-prereg-v1c.png   |
| Figure | images/20251104_123414_20251104_123412_assisted_echo_endpoint_distance__assisted-echo-t4-prereg-v1c.png  |
| Figure | images/20251104_123415_20251104_123412_assisted_echo_assist_alignment__assisted-echo-t4-prereg-v1c.png   |
| Figure | images/20251104_123415_20251104_123412_assisted_echo_seed_stability__assisted-echo-t4-prereg-v1c.png     |
| Figure | images/20251104_123415_20251104_123412_assisted_echo_telemetry_overlay__assisted-echo-t4-prereg-v1c.png  |
| Figure | images/20251104_123415_20251104_123412_assisted_echo_work_partition__assisted-echo-t4-prereg-v1c.png     |
| JSON   | logs/20251104_123411_assisted_echo__assisted-echo-t4-prereg-v1c.json                                     |
| CSV    | logs/20251104_123412_assisted_echo_ceg_summary__assisted-echo-t4-prereg-v1c.csv                          |
| CSV    | logs/20251104_123412_assisted_echo_telemetry__assisted-echo-t4-prereg-v1c.csv                            |

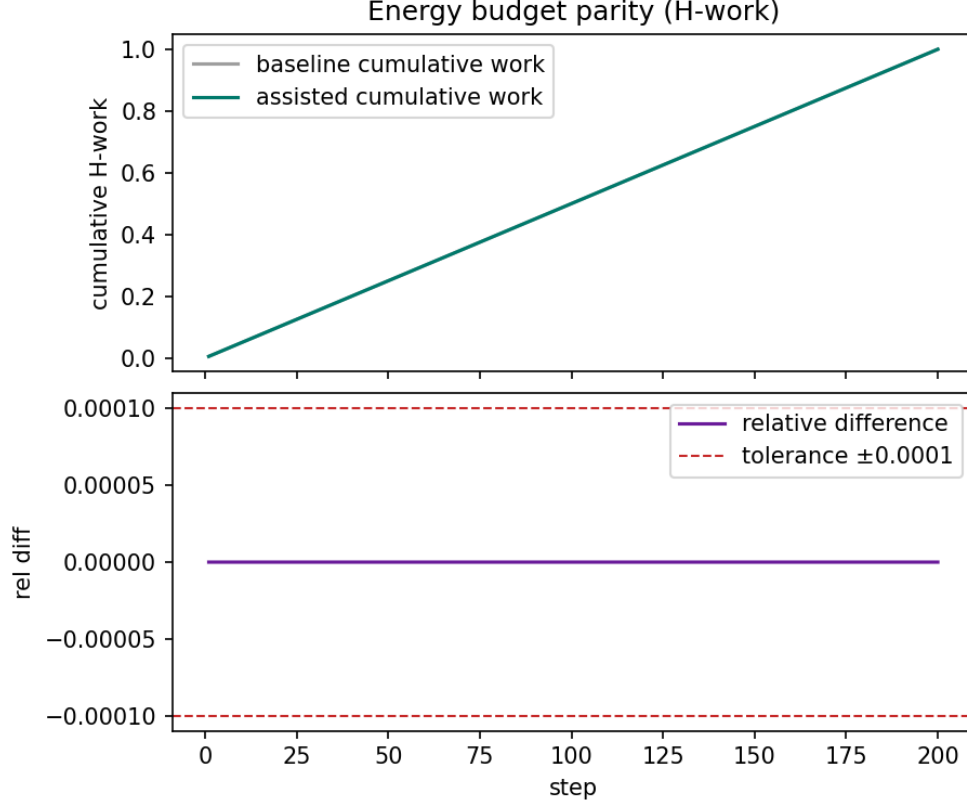


Figure 3: Energy match (reverse equal-work) baseline vs. assisted. Relative difference  $\leq 1.4 \times 10^{-16}$  (**G3 PASS**). Paired artifacts: logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json.

#### Schemas, preregistration, and specs (pointers)

- Prereg manifest: `../code/physics/metriplectic/PRE-REGISTRATION.assisted-echo-t4-prereg-v1c.json`
- Runner: `../code/physics/metriplectic/assisted_echo.py`
- Gates: `../code/physics/metriplectic/echo_gates.py`
- Metrics: `../code/physics/metriplectic/echo_metrics.py`
- Primary schema: `../code/physics/metriplectic/schemas/assisted-echo-t4-prereg-v1c.schema.json`
- Example spec (grid variants): `../code/physics/metriplectic/specs/assisted_echo.v1c_budget0p02_N512_dt0p02.json`

## 8 Discussion

The CEG curve increases monotonically across the preregistered  $\lambda$  grid, with a positive effect at  $\lambda = 0.5$  that crosses the preregistered outcome threshold. The composition accuracy (two-grid slope  $\approx 2.94$ – $2.95$  and  $R^2 \approx 0.99997$ ) supports that the observed gain is not a numerical artifact of order degradation. The energy-match gate (reverse equal-work) eliminates confounding by budget differences, so the observed signal is attributable to the model-aware correction under the instrument’s constraints.

Instrument integrity is corroborated by  $J$ -only Noether drift (G1) within scaled tolerance on all seeds and by  $M$ -limb entropy production (G2) being non-negative per step with comfortable margin. These, together with G4, indicate good internal validity for the numerical instrument at the reported settings.

Effect size is modest but robust ( $\sim 5.46\%$  median reduction at  $\lambda = 0.5$ ). Given strict energy-match and composition QC, the direction of improvement is consistent with the design of the assisted correction; larger grids and adjacent  $\lambda$  values are natural follow-ups to map the ridge of maximal gain. Learned assistance via data-driven drift/diffusion reconstruction offers an alternative model-aware correction pathway [Dat].



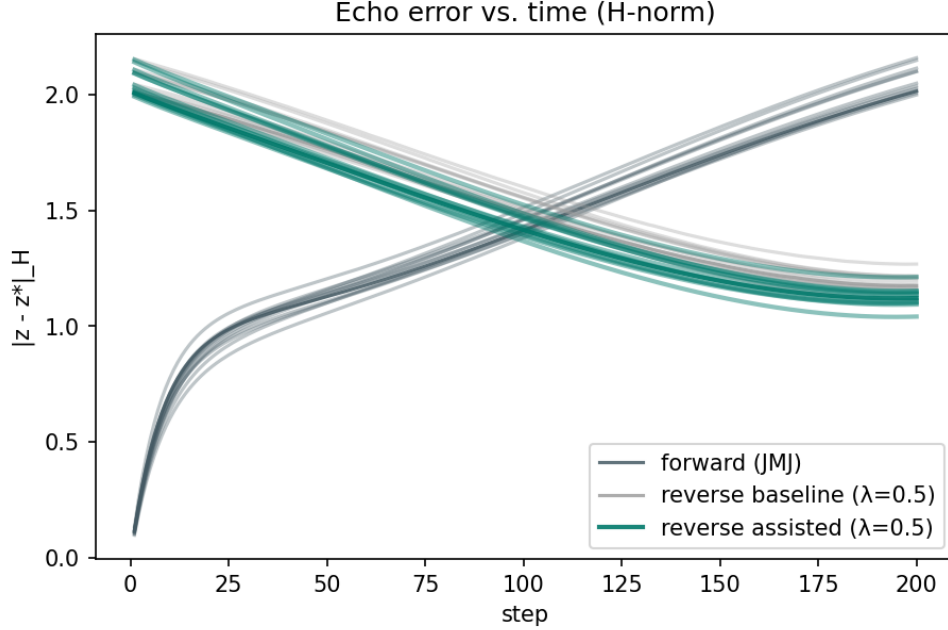


Figure 4: Error time series panels (baseline vs. assisted). Paired artifacts: logs/20251104\_123412\_assisted\_echo\_telemetry\_\_assisted-echo-t4-prereg-v1c.csv.

## 9 Limitations and Next Gates

- **Resolution and timestep sweeps:** extend to  $N \in \{512, 1024\}$  and  $\Delta t \in \{0.01, 0.04\}$  to confirm scaling of the effect under stronger accuracy demands (same gates, same thresholds).
- **Denser  $\lambda$  grid:** probe  $\lambda \in [0.3, 0.6]$  more finely to locate the peak CEG and assess curvature.
- **Ablations (RP-4):** model-blind,  $J$ -scramble,  $M$ -scramble, and  $MJM$  ordering variants to quantify dependence on internal model fidelity.
- **Multi-point  $\Delta t$  fit for G4:** confirm second-order behavior using the multi-point variant of the slope gate (same thresholds).
- **Runtime disclosure:** mine per-step timings from the run ledger for a P50/P95/P99 table and  $\beta$  slope on log-log axes (reported separately if lengthy).

## 10 Conclusions

Under preregistered conditions and strict instrument gates, the metriplectic assisted-echo exhibits a statistically positive counterfactual echo gain (median\_max 0.054552 at  $\lambda = 0.5$ ) while preserving  $J$ -Noether and  $M$ -limb monotonicity and passing Strang defect QC. Claims are bounded to these conditions and artifacts. Next gates target resolution/timestep sweeps, denser assistance controls, ablations, and expanded runtime disclosure to characterize robustness and operational cost.

## 11 Runtime and Scaling

Runtime distribution disclosure (P50/P95/P99 step time, jitter, active-site fraction, and log-log slope  $\beta$  with CIs) was not the target of this preregistered run. The run JSON ledger contains per-step timings sufficient for later aggregation. A follow-on disclosure is planned using the same seeds and configuration.

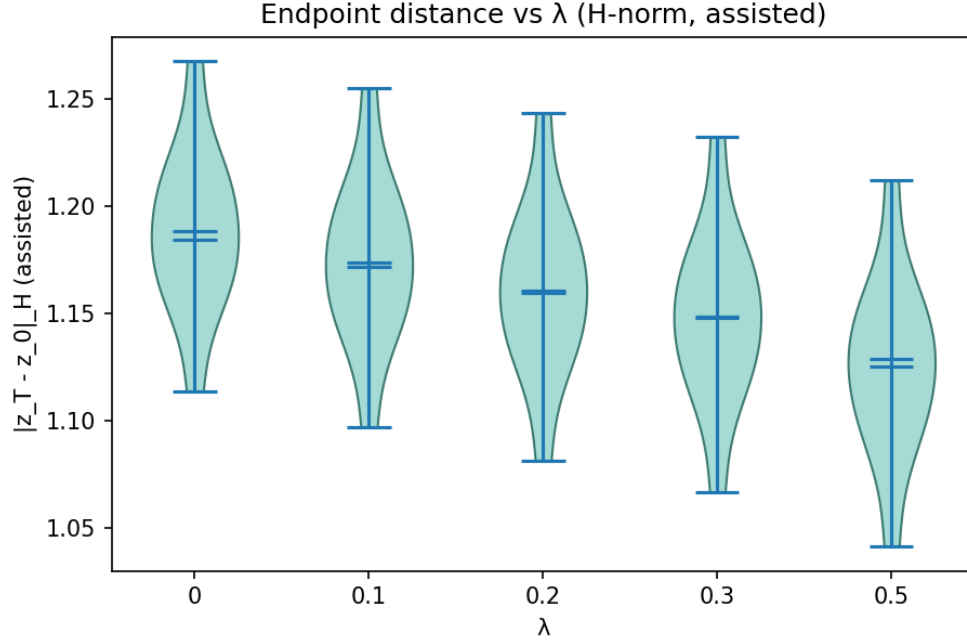


Figure 5: Endpoint distance diagnostics during reverse phase. Paired artifacts: logs/20251104\_123412\_assisted\_echo\_telemetry\_\_assisted-echo-t4-prereg-v1c.csv.

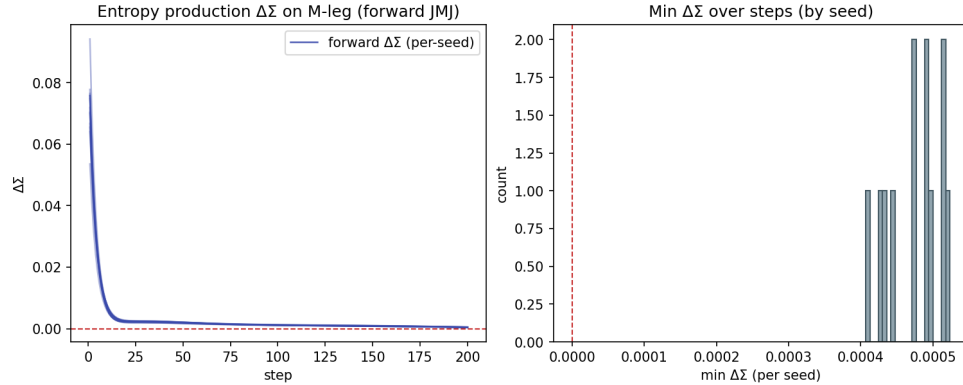


Figure 6:  $M$ -limb entropy production  $\Delta\Sigma(t)$  monotone per step; minimum per-seed  $\Delta\Sigma \gtrsim 4.1 \times 10^{-4}$  (**G2 PASS**). Paired artifacts: logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json.

Table 5: Runtime and scaling (deferred; to be aggregated from the run JSON ledger).

| Metric                  | P50 | P95 | P99 | Notes          |
|-------------------------|-----|-----|-----|----------------|
| Step time (ms)          | —   | —   | —   | to be reported |
| Active-site fraction    | —   | —   | —   | to be reported |
| Slope $\beta$ (log-log) | —   | —   | —   | CI: [ ]        |

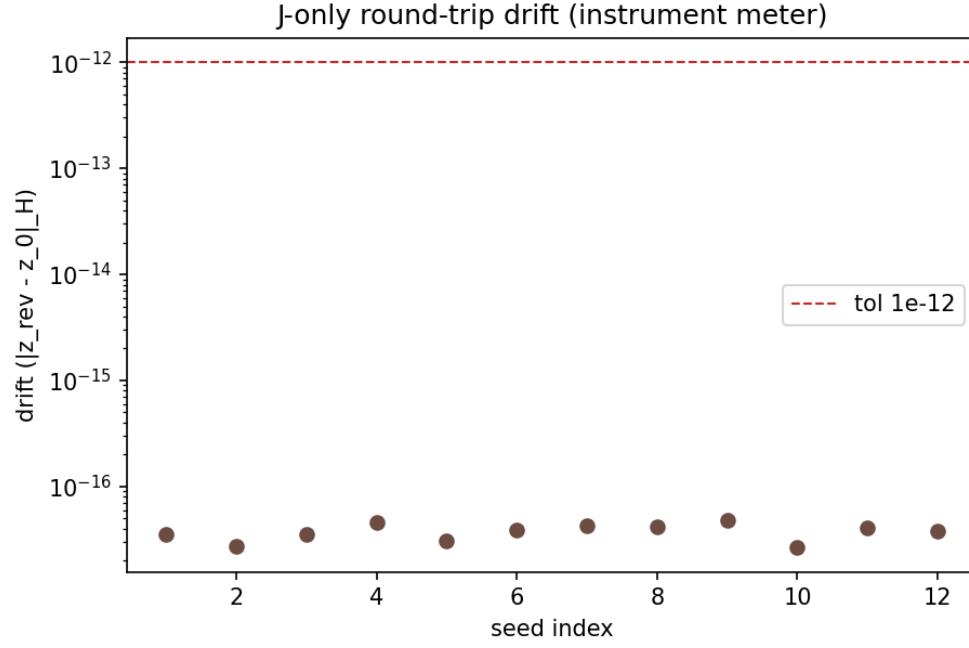


Figure 7:  $J$ -only Noether drift within scaled tolerance; typical drift  $\sim 3.5 \times 10^{-17}$  (**G1 PASS**). Paired artifacts: logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json.

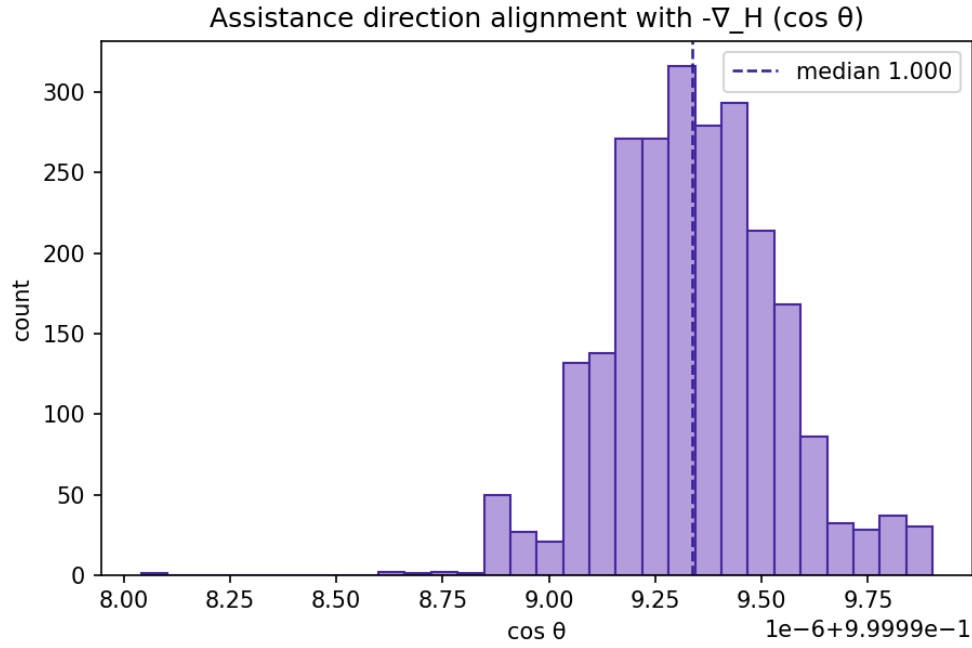


Figure 8: Assistance alignment telemetry. Paired artifacts: logs/20251104\_123412\_assisted\_echo\_telemetry\_\_assisted-echo-t4-prereg-v1c.json.

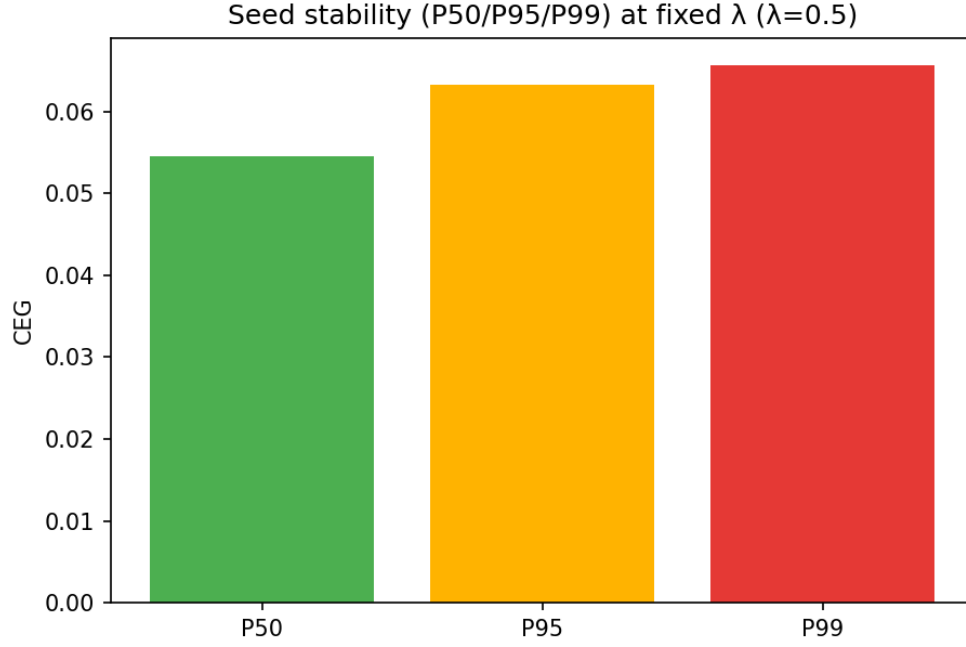


Figure 9: Seed stability summary across  $n = 12$  seeds. Paired artifacts: `logs/20251104_123411_assisted_echo__assisted-echo-t4-prereg-v1c.json`.

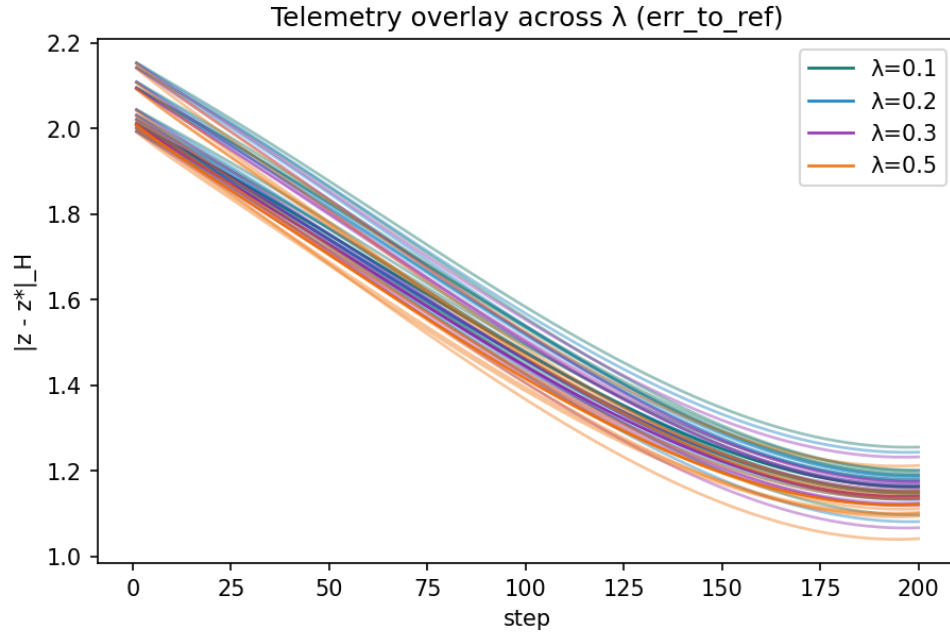


Figure 10: Telemetry overlay (forward vs. reverse traces) under equal budgets. Paired artifacts: `logs/20251104_123412_assisted_echo_telemetry__assisted-echo-t4-prereg-v1c.csv`.

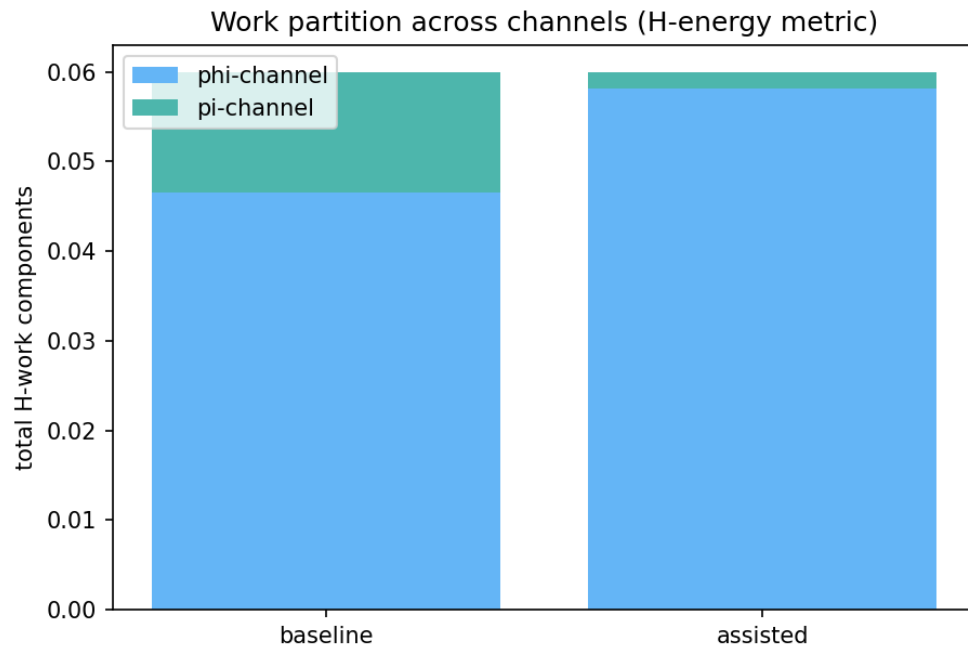


Figure 11: Work partition diagnostics (reverse phase), supporting energy-match (**G3**). Paired artifacts: logs/20251104\_123411\_assisted\_echo\_\_assisted-echo-t4-prereg-v1c.json.

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